

# 强化学习原理：动态规划方法

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2025.03.04

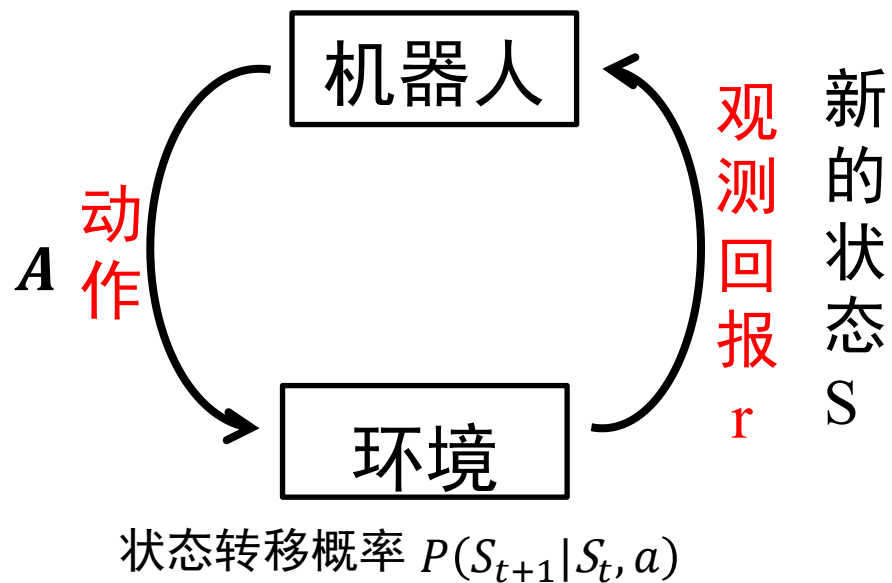
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# 强化学习的形式化表示



马尔科夫决策问题 (MDP): 四元组  $(S, A, P, r)$

策略:  $\pi: S \rightarrow u$ 。常采用随机策略:  $\pi(u|s)$

累积回报:  $R(\tau) = r_T(x_T) + \sum_{t=0}^{T-1} r_t(x_t, u_t)$

折扣回报:  $R = \sum_{t=0}^{\infty} \gamma^t r_t$

强化学习目标:  $\max_{\pi} \int R(\tau) p_{\pi}(\tau) d\tau$

值函数

值函数最优时采取的策略是最优的, 反过来, 策略最优时, 值函数最优。



# 动态规划的本质

离散贝尔曼方程：

$$\begin{aligned} J^*[x(j), j] &= \min_{u(j) \in U} \min_{\{u(j+1), \dots, u(N-1)\} \in U} \{L[x(j), u(j), j] + \sum_{k=j+1}^{N-1} L[x[k], u[k], k]\} \\ &= \min_{\substack{u(j) \in U \\ x(j+1) \in X}} \{L(x(j), u(j), j) + J^*[x(j+1), j+1]\} \\ &= \min_{\substack{u(j) \in U \\ x(j+1) \in X}} \{L(x(j), u(j), j) + J^*[f[x(j), u(j), j], j+1]\} \end{aligned}$$

求出值函数后，通过**贪婪策略**重构出最优策略

动态规划的本质是：将多阶段决策问题通过贝尔曼方程转化为多个单阶段的决策问题

# 动态规划：多阶段决策

最短路径问题：从 q 到 r 的最短路径

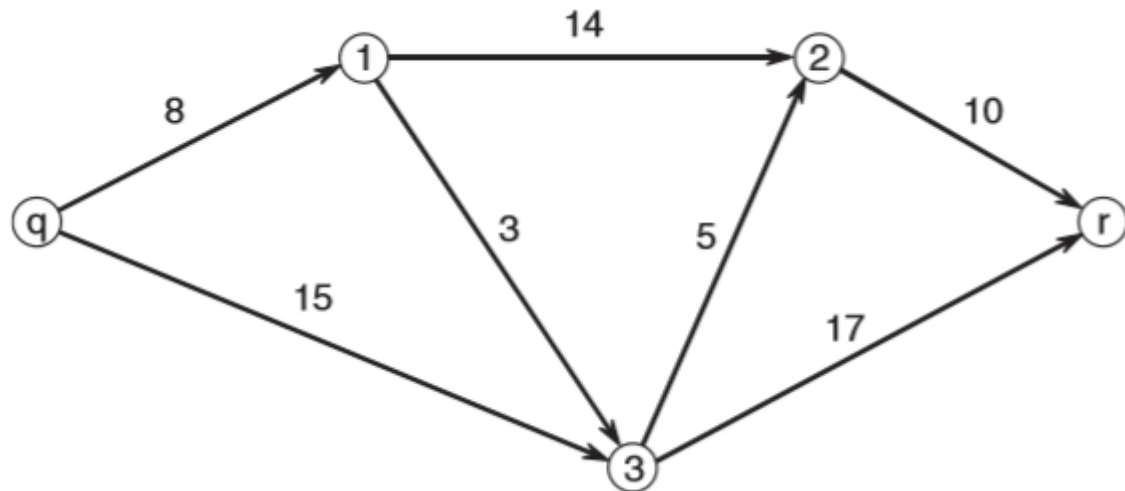


Table 1.1 Path cost from each node to node r after each node has been visited

| Iteration | Cost from Node |     |     |     |   |
|-----------|----------------|-----|-----|-----|---|
|           | q              | 1   | 2   | 3   | r |
|           | 100            | 100 | 100 | 100 | 0 |
| 1         | 100            | 100 | 10  | 15  | 0 |
| 2         | 30             | 18  | 10  | 15  | 0 |
| 3         | 26             | 18  | 10  | 15  | 0 |
| 4         | 26             | 18  | 10  | 15  | 0 |

$$J^*[x(j), j] = \min_{\substack{u(j) \in U \\ x(j+1) \in X}} \{L(x(j), u(j), j) + J^*[x(j+1), j+1]\}$$

值函数的更新公式为：

$$v_i \leftarrow \min \left\{ v_i, \min_{j \in I^+} \{c_{ij} + v_j\} \right\}$$

Warren B. Powell,  
Approximate Dynamic Programming: Solving the  
Curses of Dimensionality



# 最优控制中的动态规划

动态规划的本质是：将多阶段决策问题通过贝尔曼方程转化为多个单阶段的决策问题

连续贝尔曼方程：

$$J^*[x(t), t] = \min_{u[t, t+\Delta t]} \left\{ \min_{u[t+\Delta t, t_f]} \left[ \int_t^{t+\Delta t} L(x(\tau), u(\tau), \tau) d\tau + \int_{t+\Delta t}^{t_f} L(x(\tau), u(\tau), \tau) d\tau + \phi[x(t_f), t_f] \right] \right\}$$

$$= \min_{\substack{u(\tau) \in U \\ t \leq \tau \leq t+dt}} \left\{ \int_t^{t+dt} L[x(\tau), u(\tau), \tau] d\tau + J^*[x(t) + dx(t), t + dt] \right\} \quad (1)$$

将  $J^*[x(t) + dx(t), t + dt]$  进行泰勒展开有：

$$J^*[x(t) + dx(t), t + dt] = J^*[x(t), t] + \frac{\partial J^*[x(t), t]}{\partial x^T(t)} dx(t) + \frac{\partial J^*[x(t), t]}{\partial t} dt + \varepsilon[dx(t), dt] \quad (2)$$

将 (2) 带入 (1)，并令  $dt \rightarrow 0$  **Hamilton-Jacobi-Bellman方程**

胡寿松等，最优控制理论与系统，科学出版社，2005

$$-\frac{\partial J^*[x(t), t]}{\partial t} = \min_{u(t) \in U} \{ L[x(t), u(t), t] + \frac{\partial J^*[x(t), t]}{\partial x^T(t)} f[x(t), u(t), t] \} \xrightarrow{\text{重构}} \min_{u(t) \in U} \{ L[x(t), u(t), t] + \frac{\partial J^*[x(t), t]}{\partial x^T(t)} f[x(t), u(t), t] \}$$



# 从动态规划到强化学习

所有强化学习算法都可以看成是动态规划算法，只是用更少的计算，并且没有假设完美模型

离散贝尔曼方程：

$$J^*[x(j), j] = \min_{u(j) \in U} \min_{\{u(j+1), \dots, u(N-1)\} \in U} \{L[x(j), u(j), j] + \sum_{k=j+1}^{N-1} L[x[k], u[k], k]\}$$

$$= \min_{\substack{u(j) \in U \\ x(j+1) \in X}} \{L(x(j), u(j), j) + J^*[x(j+1), j+1]\}$$

$$= \min_{\substack{u(j) \in U \\ x(j+1) \in X}} \{L(x(j), u(j), j) + J^*[f[x(j), u(j), j], j+1]\}$$

策略改进

策略评估

连续贝尔曼方程：

策略评估

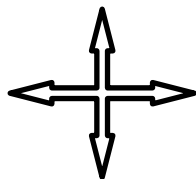
策略改进

重构

$$-\frac{\partial J^*[x(t), t]}{\partial t} = \min_{u(t) \in U} \{L[x(t), u(t), t] + \frac{\partial J^*[x(t), t]}{\partial x^T(t)} f[x(t), u(t), t]\} \xrightarrow{\text{重构}} \min_{u(t) \in U} \{L[x(t), u(t), t] + \frac{\partial J^*[x(t), t]}{\partial x^T(t)} f[x(t), u(t), t]\}$$

# 策略评估：值函数

MDP



动作

|    |    |    |    |
|----|----|----|----|
|    | 1  | 2  | 3  |
| 4  | 5  | 6  | 7  |
| 8  | 9  | 10 | 11 |
| 12 | 13 | 14 |    |

状态空间:  $S=\{1,2..14\}$

动作空间: {东, 南, 西, 北}

回报函数: -1, 直到终止状态

均匀随机策略:

$\pi(\text{东}|\cdot)=0.25$ ,  $\pi(\text{南}|\cdot)=0.25$ ,  $\pi(\text{西}|\cdot)=0.25$ ,  
 $\pi(\text{北}|\cdot)=0.25$

当智能体采用策略  $\pi$  时, 累积回报服从一个概率分布, 累积回报在状态  $s$  处的期望值定义为值函数:

$$v_{\pi}(s) = E_{\pi}[G_t | S_t = s] = E_{\pi} \left[ \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} | S_t = s \right]$$

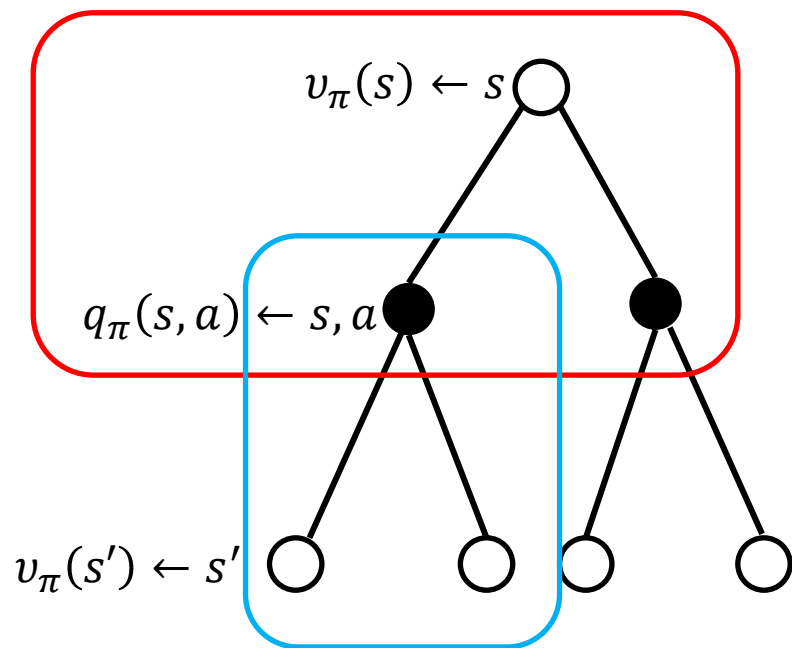
状态-行为值函数:

$$q_{\pi}(s, a) = E_{\pi} \left[ \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} | S_t = s, A_t = a \right]$$

|     |     |     |     |
|-----|-----|-----|-----|
| 0.0 | -14 | -20 | -22 |
| -14 | -18 | -20 | -20 |
| -20 | -20 | -18 | -14 |
| -22 | -20 | -14 | 0.0 |

# 策略评估(policy evaluation)

给定策略  $\pi$  构造值函数:



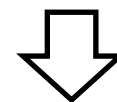
$$v_{\pi}(s) = \sum_{a \in A} \pi(a|s) q_{\pi}(s, a)$$

$$q_{\pi}(s, a) = R_s^a + \gamma \sum_{s' \in S} P_{ss'}^a v_{\pi}(s')$$



$$v_{\pi}(s) = \sum_{a \in A} \pi(a|s) \left( R_s^a + \gamma \sum_{s' \in S} P_{ss'}^a v_{\pi}(s') \right)$$

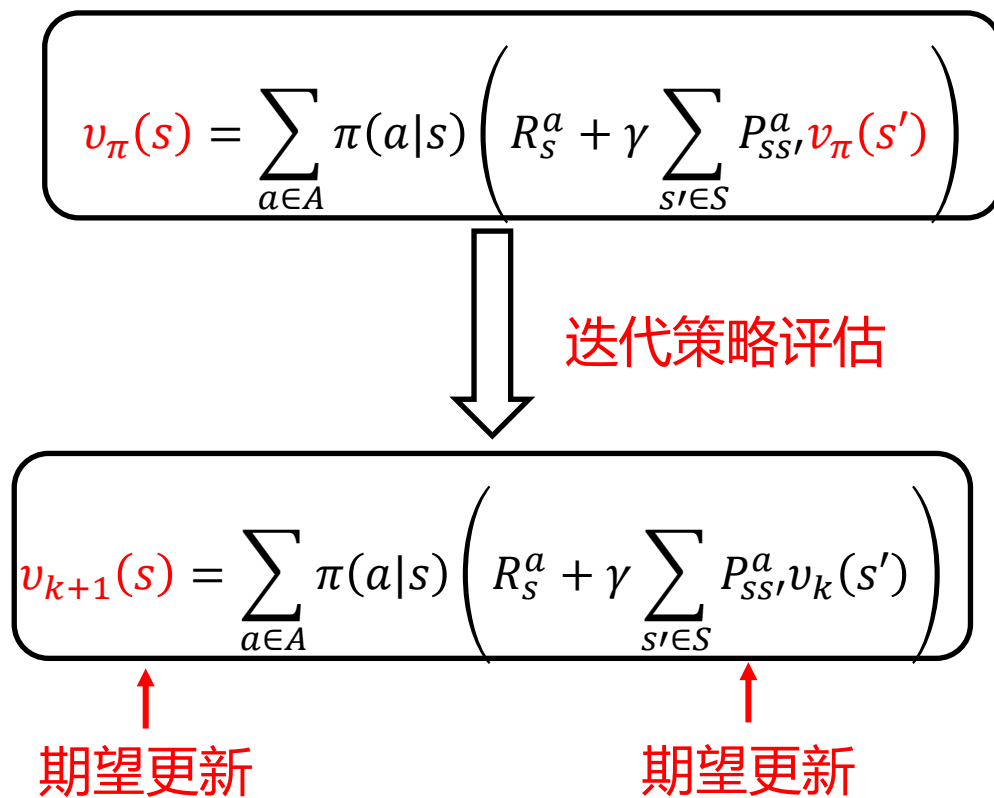
模型已知，方程组中只有值函数是未知数，方程组是线性方程组。未知数的数目等于状态的数目。



采用数值迭代算法



# 策略评估(policy evaluation)



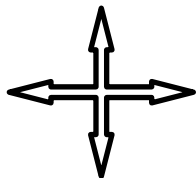
## 策略评估算法

- [1] 输入：需要评估的策略  $\pi$  状态转移概率  $P_{ss'}^a$ , 回报函数  $R_s^a$  , 折扣因子  $\gamma$
- [2] 初始化值函数:  $V(s) = 0$
- [3] Repeat  $k=0,1,\dots$
- [4]    for every  $s$  do
- [5]      $v_{k+1}(s) = \sum_{a \in A} \pi(a|s) \left( R_s^a + \gamma \sum_{s' \in S} P_{ss'}^a v_k(s') \right)$
- [6]    end for
- [7]    Until  $v_{k+1} = v_k$
- [8] 输出:  $v(s)$

一次状态扫描

# 策略评估(policy evaluation)

MDP



动作

|    |    |    |    |
|----|----|----|----|
|    | 1  | 2  | 3  |
| 4  | 5  | 6  | 7  |
| 8  | 9  | 10 | 11 |
| 12 | 13 | 14 |    |

状态空间:  $S=\{1,2..14\}$

动作空间: {东, 南, 西, 北}

回报函数: -1, 直到终止状态

均匀随机策略:

$\pi(\text{东}|\cdot)=0.25$ ,  $\pi(\text{南}|\cdot)=0.25$ ,  $\pi(\text{西}|\cdot)=0.25$ ,  
 $\pi(\text{北}|\cdot)=0.25$

## 策略评估算法

输入: 需要评估的策略  $\pi$  状态转移概率  $P_{ss'}^a$ , 回报函数  $R_s^a$ , 折扣因子  $\gamma$

初始化值函数:  $V(s) = 0$

Repeat  $k=0,1,\dots$  一次状态扫描

for every  $s$  do

$$v_{k+1}(s) = \sum_{a \in A} \pi(a|s) \left( R_s^a + \gamma \sum_{s' \in S} P_{ss'}^a v_k(s') \right)$$

end for

Until  $v_{k+1} = v_k$

输出:  $v(s)$

# 策略评估(policy evaluation)

|     |     |     |     |
|-----|-----|-----|-----|
| 0.0 | 0.0 | 0.0 | 0.0 |
| 0.0 | 0.0 | 0.0 | 0.0 |
| 0.0 | 0.0 | 0.0 | 0.0 |
| 0.0 | 0.0 | 0.0 | 0.0 |

K=0

|      |      |      |      |
|------|------|------|------|
| 0.0  | -1.0 | -1.0 | -1.0 |
| -1.0 | -1.0 | -1.0 | -1.0 |
| -1.0 | -1.0 | -1.0 | -1.0 |
| -1.0 | -1.0 | -1.0 | 0.0  |

K=1

|      |      |      |      |
|------|------|------|------|
| 0.0  | -1.7 | -2.0 | -2.0 |
| -1.7 | -2.0 | -2.0 | -2.0 |
| -2.0 | -2.0 | -2.0 | -1.7 |
| -2.0 | -2.0 | -1.7 | 0.0  |

K=2

|     |     |     |     |
|-----|-----|-----|-----|
| 0.0 | -14 | -20 | -22 |
| -14 | -18 | -20 | -20 |
| -20 | -20 | -18 | -14 |
| -22 | -20 | -14 | 0.0 |

$K = \infty$

## 算法2: 策略评估算法

输入: 需要评估的策略  $\pi$  状态转移概率  $P_{ss'}^a$ , 回报函数  $R_s^a$ , 折扣因子  $\gamma$

初始化值函数:  $V(s) = 0$

Repeat  $k=0,1,\dots$  一次状态扫描

for every  $s$  do

$$v_{k+1}(s) = \sum_{a \in A} \pi(a|s) \left( R_s^a + \gamma \sum_{s' \in S} P_{ss'}^a v_k(s') \right)$$

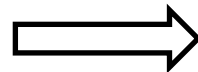
end for

Until  $v_{k+1} = v_k$

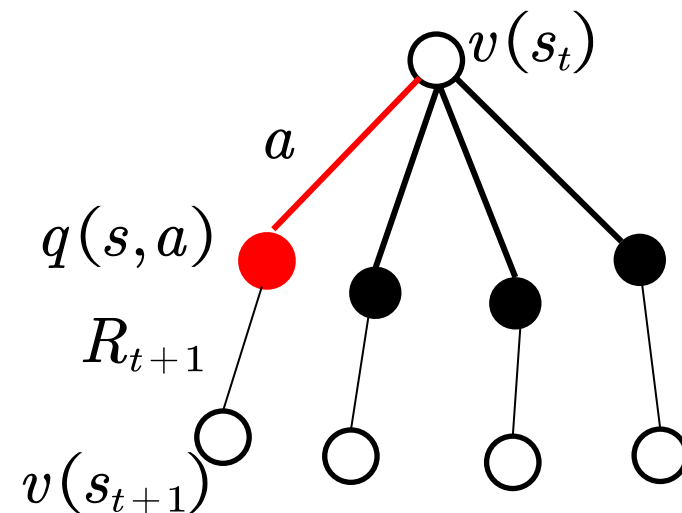
输出:  $v(s)$

# 策略改进(policy improvement)

|     |     |     |     |
|-----|-----|-----|-----|
| 0.0 | -14 | -20 | -22 |
| -14 | -18 | -20 | -20 |
| -20 | -20 | -18 | -14 |
| -22 | -20 | -14 | 0.0 |

?  
 更好的策略  $\pi_1$

$\pi_0$  均匀策略:



$$q_{\pi}(s, a) = \mathbb{E}[R_{t+1} + \gamma v_{\pi}(s_{t+1}) | s_t = s, a_t = a]$$

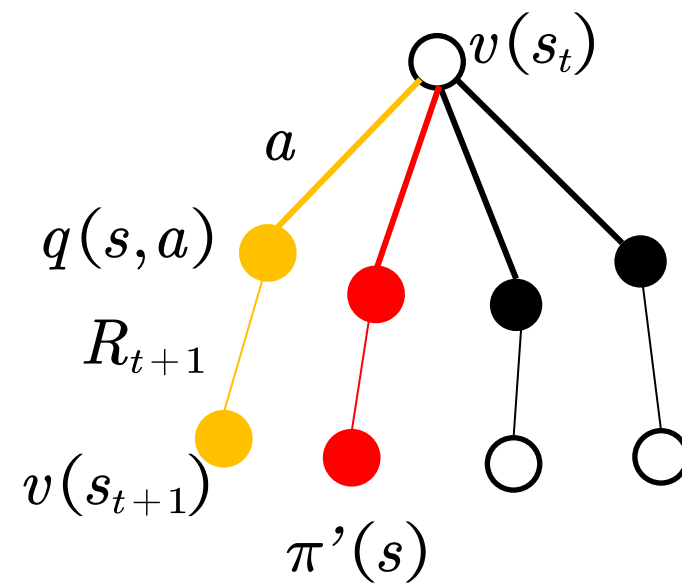
策略改善理论:

如果:  $q_{\pi}(s, \pi'(s)) \geq v_{\pi}(s)$

那么:  $\pi'$  不比  $\pi$  差, 甚至比之还要好

# 策略改进证明(policy improvement)

$$\begin{aligned}
 v_{\pi}(s) &\leq q_{\pi}(s, \pi'(s)) \\
 &= \mathbb{E}[R_{t+1} + \gamma v_{\pi}(s_{t+1}) | s_t = s, a_t = \pi'(s)] \\
 &= \mathbb{E}_{\pi'}[R_{t+1} + \gamma v_{\pi}(s_{t+1}) | s_t = s] \\
 &\leq \mathbb{E}_{\pi'}[R_{t+1} + \gamma q_{\pi}(s_{t+1}, \pi'(s_{t+1})) | s_t = s] \\
 &= \mathbb{E}_{\pi'}[R_{t+1} + \gamma \mathbb{E}_{\pi'}[R_{t+2} + \gamma v_{\pi}(s_{t+2}) | s_{t+1}, a_{t+1} = \pi'(s_{t+1})] | s_t = s] \\
 &= \mathbb{E}_{\pi'}[R_{t+1} + \gamma R_{t+2} + \gamma^2 v_{\pi}(s_{t+2}) | s_t = s] \\
 &\leq \mathbb{E}_{\pi'}[R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \gamma^3 v_{\pi}(s_{t+3}) | s_t = s] \\
 &\vdots \\
 &\leq \mathbb{E}_{\pi'}[R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \gamma^3 R_{t+4} + \dots | s_t = s] \\
 &= v_{\pi'}(s)
 \end{aligned}$$



最直观的一个改进策略是什么？

贪婪策略！

# 策略改进(policy improvement)

计算策略值的目的是为了帮助找到更好的策略，在每个状态采用贪婪策略。

$$\pi_{l+1}(s) \in \underset{a}{\operatorname{argmax}} q^{\pi_l}(s, a)$$

$\pi_0$  均匀策略: 

$\pi_1$  贪婪策略:

$K=10$

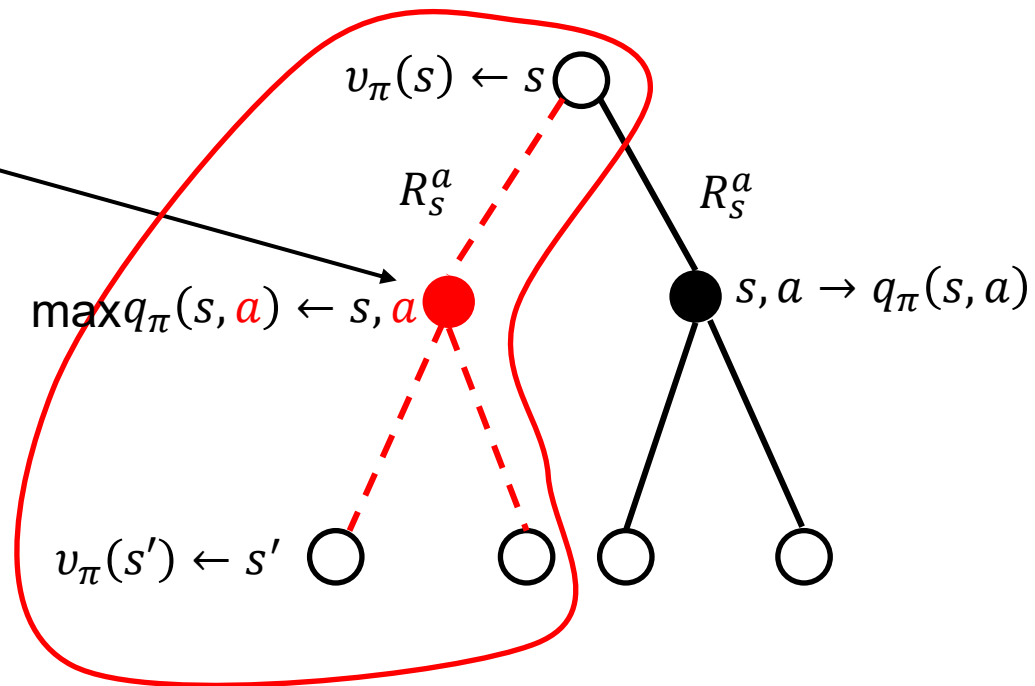
|      |      |      |      |
|------|------|------|------|
| 0.0  | -6.1 | -8.4 | -9.0 |
| -6.1 | -7.7 | -8.4 | -8.4 |
| -8.4 | -8.4 | -7.7 | -6.1 |
| -9.0 | -8.4 | -6.1 | 0.0  |

|     |   |   |     |
|-----|---|---|-----|
| 0.0 | ← | ← | ↖   |
| ↑   | ↖ | ↖ | ↓   |
| ↑   | ↖ | ↗ | ↓   |
| ↖   | → | → | 0.0 |

$K = \infty$

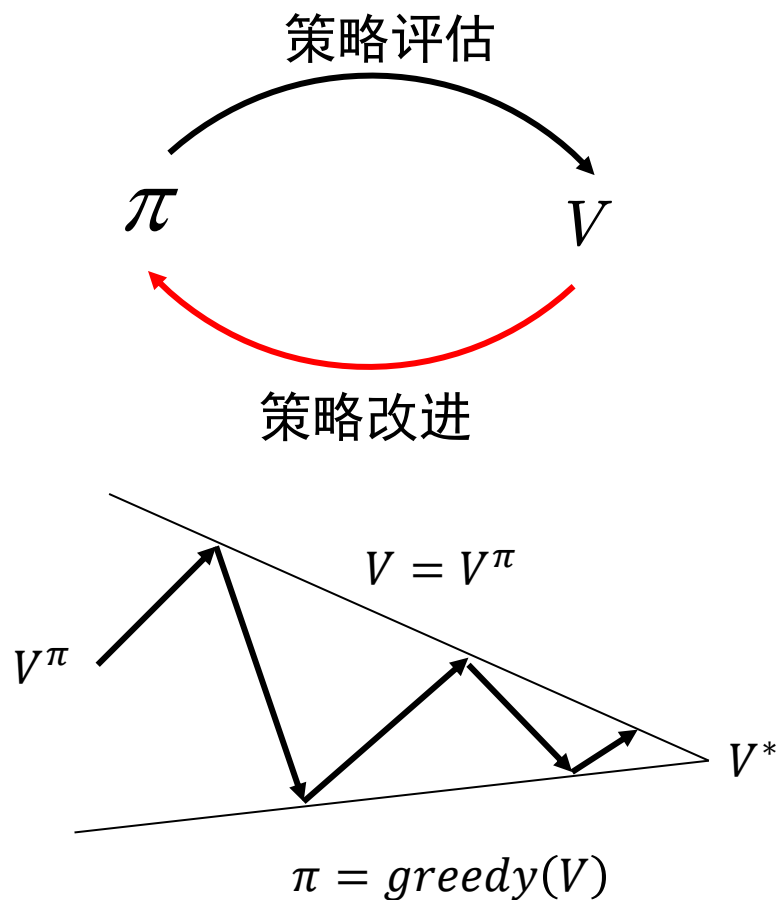
|     |     |     |     |
|-----|-----|-----|-----|
| 0.0 | -14 | -20 | -22 |
| -14 | -18 | -20 | -20 |
| -20 | -20 | -18 | -14 |
| -22 | -20 | -14 | 0.0 |

|     |   |   |     |
|-----|---|---|-----|
| 0.0 | ← | ← | ↖   |
| ↑   | ↖ | ↖ | ↓   |
| ↑   | ↖ | ↗ | ↓   |
| ↖   | → | → | 0.0 |



$$q_{\pi}(s, a) = R_s^a + \gamma \sum_{s' \in S} P_{ss'}^a v_{\pi}(s')$$

# 策略迭代(policy iteration)



## 算法1：策略迭代算法

- [1] 输入：状态转移概率  $P_{ss'}^a$ , 回报函数  $R_s^a$  , 折扣因子  $\gamma$   
初始化值函数：  $V(s) = 0$  初始化策略  $\pi_0$
- [2] Repeat  $l=0,1,\dots$
- [3]     find  $V^{\pi_l}$  Policy evaluation
- [4]      $\pi_{l+1}(s) \in \arg \max_a q^{\pi_l}(s, a)$  Policy improvement
- [5]     Until  $\pi_{l+1} = \pi_l$
- [6] 输出：  $\pi^* = \pi_l$

# 值函数迭代

策略改进一定要等到值函数收敛吗？

$\pi_0$  均匀策略: 

$K=10$

|      |      |      |      |
|------|------|------|------|
| 0.0  | -6.1 | -8.4 | -9.0 |
| -6.1 | -7.7 | -8.4 | -8.4 |
| -8.4 | -8.4 | -7.7 | -6.1 |
| -9.0 | -8.4 | -6.1 | 0.0  |

$\pi_1$  贪婪策略:

|     |   |   |     |
|-----|---|---|-----|
| 0.0 | ← | ← | ↖   |
| ↑   | ↖ | ↖ | ↓   |
| ↑   | ↙ | ↘ | ↓   |
| ↙   | → | → | 0.0 |

$K = \infty$

|     |     |     |     |
|-----|-----|-----|-----|
| 0.0 | -14 | -20 | -22 |
| -14 | -18 | -20 | -20 |
| -20 | -20 | -18 | -14 |
| -22 | -20 | -14 | 0.0 |

|     |   |   |     |
|-----|---|---|-----|
| 0.0 | ← | ← | ↖   |
| ↑   | ↖ | ↖ | ↓   |
| ↑   | ↙ | ↘ | ↓   |
| ↙   | → | → | 0.0 |

当 $K=1$ 时便进行策略改进，得到值函数迭代算法

$$v^*(s) = \max_a R_s^a + \gamma \sum_{s' \in S} P_{ss'}^a v_*(s')$$

[1] 输入: 状态转移概率  $P_{ss'}^a$ , 回报函数  $R_s^a$ , 折扣因子  $\gamma$

初始化值函数:  $v(s) = 0$  初始化策略  $\pi_0$

[2] Repeat  $l=0, 1, \dots$

[3] for **every**  $s$  do

[4] 
$$v_{l+1}(s) = \max_a R_s^a + \gamma \sum_{s' \in S} P_{ss'}^a v_l(s')$$

[5] Until  $v_{l+1} = v_l$

[6] 输出: 
$$\pi(s) = \operatorname{argmax}_a R_s^a + \gamma \sum_{s' \in S} P_{ss'}^a v_l(s')$$





# 作业



1. 阅读 Sutton 书第三章， 并做读书笔记
2. 预习策略迭代和值迭代的代码